Prime Numbers and the Fundamental Theorem of Arithmetic

- **Exercise 1**
Find all primes p such that $p+1$ is prime.
    - **Solution 1.1**
    Suppose p is a prime, then $p>1$. The only even prime is 2 because if $2 \mid p \implies p=2k$ for some $k \in \mathbb{N}$. If $k=1$, then $p=2$, and if $k \neq 1$, then $p \neq 2$ while $2 \mid p$.
    If $p=2$, then $p+1=3$ is a prime. Now suppose p is a prime with $p>2$. Then p cannot be even, so p must be odd. It follows that $p+1$ is even, but since $p+1>p>2$, the integer $p+1$ cannot be prime. Therefore, the only consecutive prime numbers are 2 and 3.

- **Exercise 2**
(1) Prove that the numbers

$$100!+2, 100!+3, \cdots, 100!+100$$

are composite.
    - **Solution 2.1**
    For each $k \in \mathbb{N}$ with $2 \leqslant k \leqslant 100$, we have $k \mid 100!$. We also have $k \mid k$, so $k \mid (100!+k)$. Since $k \neq 1$ and $k \neq 100!+k$, the integer $100!+k$ has a positive divisor different from 1 and itself. This means it is composite. Hence, $\forall k \in \mathbb{N}$ with $2 \leqslant k \leqslant 100$, the number $100!+k$ is composite.

(2) Let $N \in \mathbb{N}$. Prove that there exist N consecutive composite numbers.
    - **Solution 2.2**
    For each $N \in \mathbb{N}$, the following N numbers

$$(N+1)!+2, (N+1)!+3, \cdots, (N+1)!+N+1$$

    are consecutive composite numbers.

- **Exercise 3**
Suppose $n \in \mathbb{N}$, and p is a prime number. Prove that n^2 is divisible by p if and only if n is divisible by p.
    - **Solution 3.1**
    ($\Longrightarrow$) Suppose $p \mid n^2$. By Corollary 1.4(1), since $n^2=n \cdot n$, we have $p \mid n \vee p \mid n \implies p \mid n$.
    ($\Longleftarrow$) If $p \mid n \implies \exists k \in \mathbb{N}$ such that $n=kp$, then $n^2=k^2p^2=p\left(k^2p\right) \implies p \mid n^2$.

- **Exercise 4**
Prove that a positive integer is a perfect square if and only if every exponent in its prime factorization is even.
    - **Solution 4.1**
    ($\Longrightarrow$) Suppose $n \in \mathbb{N}$ and there exists $a \in \mathbb{N}$ such that $n=a^2$. Consider the prime factorization of a. By the fundamental theorem of arithmetic, there exist distinct prime numbers $p_1, p_2, \cdots, p_r$ and exponents $\alpha_1, \alpha_2, \cdots, \alpha_r \in \mathbb{N}$ such that $a=p_1^{\alpha_1}p_2^{\alpha_2}\cdots p_r^{\alpha_r}$. Then

$$\begin{aligned} n = a^2 &= \left(p_1^{\alpha_1}p_2^{\alpha_2}\cdots p_r^{\alpha_r}\right)^2 \\ &= p_1^{2\alpha_1}p_2^{2\alpha_2}\cdots p_r^{2\alpha_r}. \end{aligned}$$

    By the uniqueness of the fundamental theorem of arithmetic, this expression represents the prime factorization of n, where every exponent is even.
    ($\Longleftarrow$) Suppose the prime factorization of n has even exponents, that is, $n=p_1^{2\alpha_1}p_2^{2\alpha_2}\cdots p_r^{2\alpha_r}$. Then

we can write $n = (p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_r^{\alpha_r})^2$. Since $p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_r^{\alpha_r} \in \mathbb{N}$, n is a perfect square.

- **Example 5**
  Let x, y be two positive integers. If $x \cdot y$ is a perfect square and x itself is a perfect square, prove that y is a perfect square.
  - **Solution 5.1**
    By Exercise 4, a positive integer is a perfect square if and only if every exponent in its prime factorization is even.

    By Proposition 2.4, for any prime p, we have

    $$\nu_p(xy) = \nu_p(x) + \nu_p(y).$$

    Since xy and x are perfect squares, $\nu_p(xy)$ and $\nu_p(x)$ are even integers. Since the difference between two even integers is even, $\nu_p(y) = \nu_p(xy) - \nu_p(x)$ must be even. Thus, every exponent in the prime factorization of y is even, which implies y is a perfect square.

- **Exercise 6**
  Prove that the remainder obtained by dividing a prime number by 30 is either 1 or a prime number.
  - **Solution 6.1**
    Let p be a prime number. Dividing p by 30 according to the division algorithm yields

    $$p = q \cdot 30 + r \quad \text{with} \quad 0 \leqslant r < 30.$$

    We want to prove that $r = 1$ or r is a prime number. We consider the possible cases for r:
    - If $r = 0$, then $p = 30q$. Since p is prime, this implies $p = 30$ and $q = 1$, but 30 is not a prime number, which is a contradiction.
    - Suppose $r > 0$ and r is composite. Then r must have a prime factor s such that $s \leqslant \sqrt{r} < \sqrt{30} < 6$. Thus, $s \in \{2, 3, 5\}$. Since s is a prime factor of 30, we have $s \mid 30$, and since $s \mid r$, it follows that $s \mid (30q + r)$, which means $s \mid p$. Since p is prime, we must have $s = p$. This implies $p \in \{2, 3, 5\}$. However, if $p \in \{2, 3, 5\}$, then dividing p by 30 yields a remainder $r = p$, which is a prime number, contradicting the assumption that r is composite.

    Therefore, the remainder r cannot be 0 or a composite number, meaning it must be either 1 or a prime number.